

Course Design and Development

This summary is adapted from Ko, S., & Rossen, S. (2017). Teaching online: A practical guide. Retrieved from <http://ebookcentral.proquest.com>.

Developing and designing an online course is more than just copying your content into a digital format. To ensure your course provides learners a coherent and effective learning experience, you must be purposeful and intentional about the overall design and development of the course. Going online also requires you to consider how to leverage the tools in the Learning Management System (LMS) to optimize the learners' learning experience.

So, where can you begin? We will explore some initial steps to help you get started.

Initial Steps for Designing and Developing a Course

While you are not expected to be an expert in instructional design principles, it can be helpful to be familiar with a basic framework or process for designing online courses. The “backward design” model promoted by Wiggins & McTighe in *Understanding by Design* (1998) might prove a helpful concept. This model encourages you to design your course with the “end” in mind.

In this model, the three necessary stages of design require you to:

1. **Identify the desired results or objectives:** In this stage, consider what the goals or objectives are for a course or module. Instructors consider what knowledge students should be familiar with, what knowledge students should master, and what knowledge students will retain long after a course is complete. This third piece of knowledge is often referred to as enduring understandings in the “backward design” model as is often adapted to become the desired results or objectives of the course.
2. **Determine acceptable evidence:** In this stage, consider the activities and assessments students will complete in order to demonstrate evidence of understanding and learning. Assessments may include papers, quizzes, lab projects, group projects, presentations, etc.
3. **Plan learning experiences and instructions:** In this stage, you determine what readings, videos, or activities you will include in the course to provide students the skills and knowledge needed to achieve the outcomes. You can use the [Understanding by Design \(UbD\) template](#) to plan your course

or module. To download the template, click on Design Tools. Then, click on UbD Unit Template (version 2.0) from the drop-down list.

In addition to the objectives, learning activities, and assessment, it's also important to plot out the basic structure of your course to ensure it is clear and easy to navigate. When possible, online courses should be fully built prior to the start of the course. Building your online course while you are teaching will likely not result in a well-integrated and cohesive learning experience. Teaching while building your course can also contribute to the stress of teaching online.

With this basic approach in mind, let's look at three elements of converting to or creating an online course:

- Analysis
- Course goals and learning objectives
- Design

Analysis

Several instructional design frameworks include an analysis phase. In this phase, you might analyze whom your course is for, how it fits in with the program curriculum, what the course learning objectives are, and what resources will be available to you and your students.

Here are some of the questions Ko & Rossen suggest you ask during the analysis phase (p. 43):

1. Who is your student audience? For example, is this course for 20 or 100+ students? Is the course for beginners or advanced students? For majors or primarily non-majors? What is the average age range of students and the percentage of working adults? Are most students local or do some reside at a considerable distance? What existing knowledge might students have on this topic?
2. Will new students have an orientation to how to succeed online prior to taking your course? How much information will you need to provide to manage expectations?
3. What support will you have available to assist you in creating online course materials?

Establishing the instructional goals and objectives of your course is an integral component of the analysis phase. Let's take a closer look at the difference between goals and objectives, and one approach to developing objectives.

Course Goals and Learning Objectives

If you're wondering what the difference between goals and objectives are, you're not alone. In their book, Ko & Rossen explain that "the difference between goals and objectives is basically the difference between things that can be known but not easily measured and those that can be demonstrated" (Ko & Rossen, 2017, p. 42). At some universities, goals may be expressed as competencies. They tend to be more general than objectives and, oftentimes, describe a skill required for a job or profession.

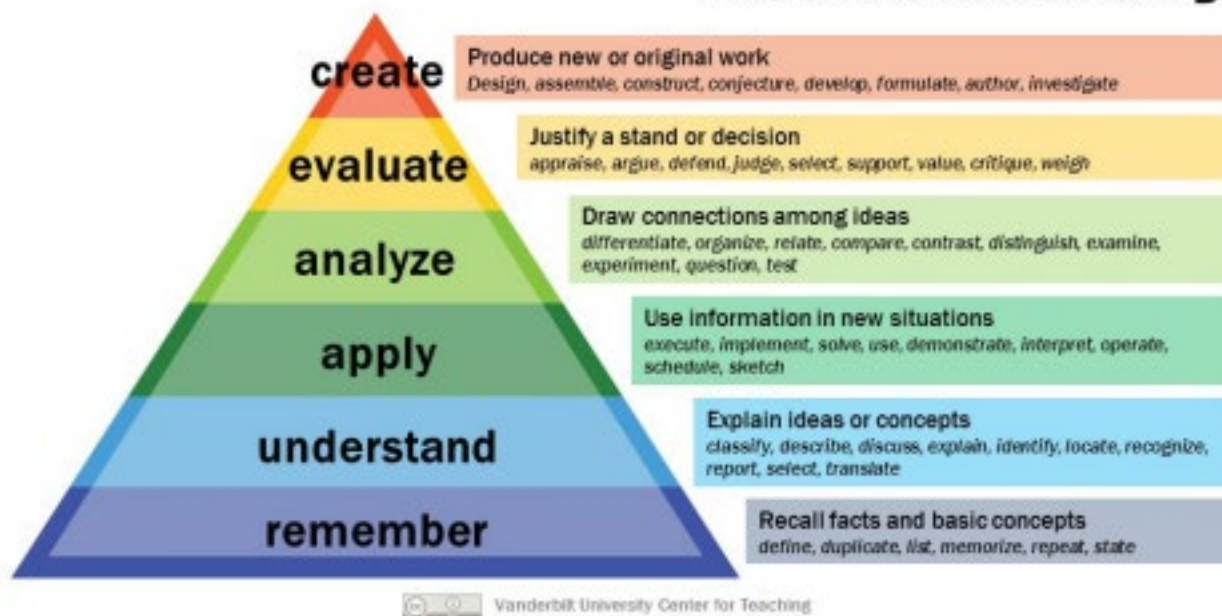
Objectives, on the other hand, are clear, descriptive statements that use measurable verbs to articulate what you want students to know and be able to do upon successful completion of your course. In addition to communicating to your students what is expected of them in the course, learning objectives can help you focus on what is to be taught and learned so you can plan accordingly. Learning objectives can help you select and organize course content and learning activities and determine appropriate assessments that fit the needs of your course.

However clever, amazing, or interesting a reading, resource, or activity may be, if it doesn't help your students achieve the stated learning outcomes, you may want to make it optional or remove it from the course altogether.

Rather than writing learning objectives for every activity, you might start by concentrating on the overall course objectives and how they can be applied for one or more units or modules of the class. Consider avoiding verbs like know, learn, and understand for your objectives. These are difficult to measure and belong in the realm of *goals*. Another tip is to keep overall course learning objectives to a minimum to help you keep you focused on the essentials.

One approach to developing learning objectives is to use Bloom's Taxonomy, a hierarchical system of classifying different levels of thinking. There are numerous charts and graphic representations of Bloom's Taxonomy.

Bloom's Taxonomy



Here are some additional resources on Bloom's Taxonomy:

- [Integrating Technology with Bloom's Taxonomy - Arizona State University](#)
- [Revised Bloom's Taxonomy - Iowa State University](#)

Ko & Rossen share the following quote on writing objectives from Sharon Guan, an experienced instructional designer and Director of Faculty Instructional Technology Services at DePaul University:

Sometimes objectives are very subtle in the faculty's mind and if a faculty member has no designer to work with, that faculty member should just ask herself, "What do you want students to be able to do as a result of this class and how would you know that students have learned or know how to do this?" (2017, p. 46)

The book goes on to share Sharon's thoughts on how "it is possible for someone to write very precise learning objectives and yet have a poorly designed course" (2017, p. 46). It is not enough to just write clear, measurable learning objectives. Instructional materials and assessments should clearly align to those objectives.

Note: If the learning objectives for the course you are teaching have already been provided, consider focusing your time on designing learning activities that will align with those objectives.

Course Design

With your learning objectives set, you are ready to focus on the design of your course. “Design” refers to the “shape and direction you want your course to take” (Ko & Rossen, 2017, p.47). When you consider the design of the course, take into consideration your course learning objectives, your preferred teaching strategies and approaches, and any new approaches or educational technology you would like to use in the online environment.

According to Ko & Rossen, there are two main design principles to keep in mind:

1. Make sure the course learning objectives are defined in terms of the learning outcomes you want students to be able to demonstrate.
2. Make sure to align all activities, assignments and assessments with your stated learning outcomes.

As you begin to design your course, here are some of the questions Ko & Rossen suggest you ask yourself. These are also good questions to discuss with an instructional designer if you have one available to you.

1. How might you re-envision some of your face-to-face activities for the online format?
2. Is group work or peer review work appropriate or desirable for your course?
3. What is the best way to assess student learning in the context of your course or discipline?
 - a. Portfolios?
 - b. Quizzes or exams?
 - c. Essays or research papers?
 - d. Fieldwork reports or other experiential learning documentation?
 - e. Individual projects and/or presentations?
4. What will be the balance of student-centered versus instructor led activities? Will you mainly facilitate discussion and research, or does the course have a strong component of lecturing and instructor commentary as well?
5. How central is discussion or student presentation to achieving the objectives of the course? Do you want to take advantage of the many new tools that would allow students to generate content in text, audio, or video?
6. What are your preferred methods of presenting content?

- a. Do you have graphics or slides that you want to utilize in some way?
 - b. Do you use lecture notes or commentary and will these be converted or substantially revamped?
 - c. Is it important for you to accommodate as many different approaches (visual, auditory, project based, collaborative) to learning as possible?
7. How might your available resources affect the implementation of your design?
 - a. Do you have online testing capabilities? If so, would these be mainly for self-assessment or do they possess the necessary security for graded tests?
 - a. What forms of communication are options—for example, do you have a discussion board, a chat or other synchronous software?
 - b. Is audio or video an option for presentation or activities in your online classroom and do you have the necessary tools to create such media? Will you need some training or support?

Note: As you design your course for the online environment, you have an opportunity to try out new methods, approaches, and technology. Maintaining the integrity and quality of your course does not mean that you have to approach the course the same way you do in your face-to-face class.

Assessing the Quality of Online Courses

There are a number of rubrics and guidelines that have been developed to guide and evaluate the quality of online courses. [Quality Matters](#)™ is a widely adopted system for design and evaluation of online courses as well as a sophisticated training and peer review program. The rubric is available to subscribing institutions or individuals, such as Arizona State University.

Other rubrics and guidelines that you may find helpful as you design and develop your course include the following:

- California State University, Chico has a Rubric for [Exemplary Online Instruction \(EOI\)](#).
- The [Open SUNY Course Quality Review \(OSCQR\) Rubric and Process](#) was developed by Open SUNY Center for Online Teaching Excellence (COTE) to address instructional design and accessibility of online courses.

Other Design Considerations

Universal Design and Accessibility

Oftentimes when we think of accessibility, we think of how to accommodate students with disabilities. However, it would be better to think in terms of universal design in learning (UDL).

Universal design in learning means to design course content and instructional activities that can be utilized by the greatest number of students to the greatest extent possible without having to make special adaptations. Additionally, universal design for learning principles are informed by the idea of providing multiple paths, perspectives, and approaches to learning (Ko & Rossen, 2017, p. 52).

Consider video captions as an example. Not only do they help students who are hearing impaired, they can help students who find it easier to read the transcript.

When designing and developing your course, consider creating content that is easily readable by eyes and screen reading software. Here are some basic tips that Ko & Rossen (2017) suggest:

- Create documents using 12-point font and use header styles to ensure consistency and ease of scanning by students using screen readers
- Create text descriptions (alt-text) for all images that describe the appearance and function of an image. This ensures that visually impaired users are presented with the same content. Also, if an image file can not load, alt text will be displayed in place of the image.
- Create captions or transcripts for all video and audio files. Many hosting services, such as YouTube, have auto caption features.
- Create meaningful titles or descriptions for all hyperlinks. The link text should describe what students will see when they click on it. Avoid posting the full URL or creating link titles that say "here" or "click here".
- Test out all external sites and tools to ensure they are accessible. Be prepared to provide alternate sites or assignments when they are not accessible to all.

Designing your course to be as accessible as possible from the beginning can help save you time later. Here are some additional resources you might consult when designing your course:

- [Universal Design for Learning \(UDL\)](#), Colorado State University

- [Accessibility and Universal Design in Learning](#), CUNY School of Professional Studies at

Adult Learners

Since many of the students enrolled in online programs are working adults, there are some key adult learning principles you might integrate into both the design and delivery of your course.

Here are some tips that Ko & Rossen (2017) suggest:

- Adult learners tend to be self-directed and independent, although they still benefit from guidance, examples, and feedback from instructors and peers. Organize your course to include ample feedback intervals, opportunities to engage with peers, and activities that facilitate student exploration and discovery.
- Adult learners like to draw on their experience and relate what they learn to real world applications. Assignments should allow for reflection, application, active learning, and sharing of ideas with others.
- Adult learners have varied levels of preparation so include just-in-time resources to assist them in improving their skills or background knowledge.

These principles are beneficial for any course but are particularly valuable when your students are adult learners.

Scaffolding

Scaffolding an assignment means to “incrementally arrange the tasks (or a series of assignments) so as to allow students to gradually build on their knowledge and skills and to provide the necessary support resources at each stage (Ko & Rossen, 2017, pp. 54-55).

For example, a final paper can be broken down into smaller assignments— topic selection, annotated bibliography, first draft, before the final paper is due. This is particularly helpful in online courses that are condensed as it allows for students to ask questions and receive feedback.

Scaffolding can be used in any subject matter or discipline although it can be especially an effective way to demonstrate the proper steps or procedures in certain disciplines.

Course Development

Having assembled your materials and analyzed the needs of your class, with your course objectives and basic design considerations in mind, you are ready to develop your course. The development stage involves the actual creation of the course, including the learning materials, activities and assessments.

Getting Organized

To help you stay organized and ensure alignment between your course learning objectives and your course content, you might consider using a course planning worksheet, which is simply a table that provides a general outline of the major units in your course and their topics, the learning materials (e.g. readings, videos, etc.) and assessments (e.g. discussions, quizzes, etc.).

Using a course planning worksheet can help give you a holistic view of the course to ensure you are providing the learning materials students need to achieve the stated learning outcomes. The worksheet can also ensure that you sequence your assignments and activities into a logical order. “Whether your units comprise one week or several, you will need to work out the sequence in terms of weekly units, which is generally how students approach an asynchronous online class” (Ko & Rossen, 2017, p. 57).

As Ko & Rossen explain, the learning content and activities in most college classes can probably be divided into a few large categories:

- Instructor-generated content and presentations (e.g. lectures, simulation, charts)
- External content (e.g. textbook, articles, guest lectures)
- Course Discussions (synchronous or asynchronous)
- Group work or student-created content (e.g. wikis, peer review, group projects)
- Research (e.g., experiments, fieldwork, interviews)
- Assessments (e.g. exams, essays, projects, papers)

Let's review some of the factors you need to consider in converting these various types of course activities to an online format.

Instructor-Generated Content and Presentations

Lectures are probably the most common method of presenting content in a traditional college classroom. There are several ways you can translate your lectures for the online environment:

Narrated Slides, Screencasts, or Video Productions. Narrated slides or screencasts can be an effective way to present materials online. However, avoid simply narrating bulleted text.

Here are some tips for using narrated slides effectively online:

- Use multiple slides to describe a concept rather than lecturing on just one slide.
- Use bullet points rather than complete sentences so students spend their time listening to you rather than reading the slide.
- Use good slide design. For example, limit the number of fonts and colors you use and be consistent throughout.
- Use instructionally effective visuals, graphics, arrows, video snippets, etc. to reinforce the content.
- Use a conversational voice to narrate your slides and show your interest and passion for the topic.

A narrated slide presentation can be helpful when walking students through a series of steps—say, the steps for creating a business plan. Audio alone or in narrated slides can also be a very effective instructional method in foreign languages, art, and music. Audio can also be used to replace a short lecture or a guest lecture. Audio can also serve as a personalized introduction to the course or to the instructor. In the latter case, a short audio welcoming message might be combined with a still photo. Audio alone can be more digestible to students who are on-the-go.

More and more institutions are investing in studios where faculty can create lectures that are optimized for online delivery. We encourage you to check with your department to see what options are available to you. However, if your institution does not provide that sort of resource, you will find that it is increasingly easy to do this on your own. One advantage is that these informal videos add a human element to the course by letting your personality shine through.

In addition to lectures, effective use of videos include a demonstration of a science experiment or a presentation of vignettes for a language course. Creating these types of demonstrations allow instructors to be both focused and customized in their approach.

When offering audio and video, it is important to take into consideration students' disabilities. Consider creating captions or preparing a text transcript or summary for the benefit of those with disabilities. This also serves to give all students additional options

for accessing the material, including those who have technical problems and those for whom the text option is a learning preference.

Slideshow Presentations. If you're unable to record a video or screencast due to time or technology restrictions, you can opt to share your slide deck with students, although research indicates that students tend to find these less engaging and valuable to their learning compared to slides with video or audio narration (insert citation). Since you won't be expanding on each slide, the slides themselves must be detailed and carry the entire presentation. The design of your slides and the images and graphics you use are essential to the success of such presentations.

Keep in mind, too, that slide presentations, when they involve a large number of slides and click-throughs, can be very tedious to the viewers. If you wish to use slide-show presentations, divide up your slides and test them so that each segment takes no more than ten minutes. That will make it easier for students, not only to initially absorb, but also to find particular segments for review.

Using External Content

Instructors often rely on external content, such as textbooks and articles for their learning materials. While it is important to check textbook and ebook availability, this might be a good opportunity to see if there are any open educational resource (OER) versions of readings available. When appropriate, selecting OERs can provide considerable cost savings to students. Here are some resources that you can access for more information on OERs.

- [Open Educational Resources: the Basics \(pdf\)](#) A quick, 2-page primer on OER basics - if you take nothing else, take this.
- [7 Things You Should Know about Open Educational Resources \(pdf\)](#) From the EDUCAUSE Learning Initiative, a brief overview of Open Educational Resources
- [Finding Education / OER Resources - Creative Commons](#) Selected list of locations for finding open content and tools, along with recommended strategies, grouped by type of content.

Other types of external content, such as TedTalk videos or guest lectures are a great way to provide supplemental instruction to students and enrich the learning experience.

Discussions

Discussions are often used in the online environment to build a sense of community, to provide opportunities for students to engage with one another and their instructor, and to assess student learning. When designing a discussion, you will need to decide whether or not you want the discussion to be carried out asynchronously or synchronously. This is one area where your course analysis comes into play. If you're conducting a hybrid course, you may want to have all real-time discussion carried out in the classroom and utilize an asynchronous forum for follow-up discussions that allow for more reflection.

If your course is entirely online, your students may be living in multiple time zones and juggling work, school, and family obligations. In this case, asynchronous discussions provide the most flexibility to your students.

Asynchronous Discussions. It can be helpful to determine how you want to use discussions in relation to the other learning materials and assessments in a unit or module. For example, you may choose to have the discussion topics related to the questions you raise in your lectures and other presentations, or you may use the discussions to introduce additional materials or further application of the ideas you have presented.

Discussions can benefit any discipline, including math and science courses despite what some might say. One way to use the discussion forum in a science and math course is to have a question and answer forum where students post questions and get answers from the instructor or TA. While this can be time consuming, it allows the instructor to broach new or more in-depth topics.

Another way to use the discussion forum is to have students offer possible solutions to problems. The discussions then consist of analyzing both the process of solving a problem and the solution itself. This allows students to be exposed to multiple ways to approach a problem.

When designing the discussion, include your expectations for participation, including how often students are expected to post and how long their posts should be. One effective way to communicate your expectations and to facilitate grading is to create a discussion rubric.

Synchronous Discussion. Synchronous discussions can be text-based chat or may be done using a video conferencing tool. It may be webinar-like in that it is mostly the instructor delivering content, with students permitted to ask questions via chat or voice. Or it may be an opportunity for students to work in online groups.

In chat and all synchronous conferencing, everyone participating must be present at

the same time. The conversation takes place in real time, although your synchronous program may allow for saving the chat transcript or video so that it can be read (or listened to) after the actual event.

While synchronous sessions can be more spontaneous than asynchronous communication, adequate preparation and forethought are still essential to its success, especially if they are to be used to deliver instructional content. The timing of synchronous sessions, for example, is an important factor in planning, especially if you have students residing in different time zones.

It might be helpful to survey students to find the most convenient times for all or you may choose to set up two or more real-time meeting times and require that each student sign up for a particular time slot so that you can control the number participating in each session.

When using synchronous sessions, it is important to provide students adequate preparation time by announcing the topics and/or questions beforehand and by posting the rules for conduct during the session.

Group-Oriented Work

All types of group activities, from peer review to true cooperative learning exercises, are possible in the online environment. To be effective, however, group activities must be well organized and properly paced. Below are some considerations you might consider when planning and implementing group work.

When planning on integrating group work into your course, it is important to consider how many group activities you plan to have and whether or not you will be changing groups for each activity. In the online environment, it is helpful to provide students ample time to get to know one another and to develop a working relationship. Group organization and working procedures often take longer in the online environment.

It is also important to consider how, where, and when groups will meet and work together online. You might want to think about how you plan to monitor and evaluate individual contributions if this is an area you are concerned about.

For groups to be effective in the online environment, provide specific guidelines for working together. You might require groups to fill out a group contract, which is a document that formalizes the expectations for the group, including the frequency and duration of meetings, the behaviours expected of all group members, the process for dealing with unmet expectations or problems, the method for feedback, etc.

Student groups will also benefit from having multiple avenues for communication as possible. You can leverage the tools within your Learning Management System or

use online collaborative tools, such as Google Docs and Google Slides.

You will also want to consider how you plan to grade group work. You can assign a grade to the entire group or to each individual contributor. It can be helpful to create a grading rubric for students to see, in order to clarify your criteria for grading. You might also consider using a peer evaluation form so students can provide feedback on their peers' contributions. Whatever you choose to do, it's important to carefully design these activities.

Research

Research activities, including fieldwork, may be carried out in a number of ways in an online class. Here are some examples of research you might consider incorporating into your own course:

- **Web-Based Research:** If you have students conducting web-based research, it may be important to create a list of guidelines for evaluating and searching for credible online materials. You might also enlist the help of your campus librarians to either put together a list of pre-approved websites or to create materials on digital literacy that you can use in your course.
- **Library Research:** Depending on your university and the facilities, materials, and services that your library has available online, you can design learning activities that direct students to the online databases your school has available. Again, you can leverage your campus librarian's knowledge and expertise to find out what databases and materials would be available for your discipline.

Fieldwork or Internships

The online environment may not seem like the obvious option for fieldwork or internships since much of what students do is accomplished outside the online environment. However, you can have online students discuss their experiences, share their results, and reflect on their learning online. Online teacher-education and professional practice programs may incorporate periodic reports from the mentor or supervisor at the student teaching site, as well as videos capturing the teaching assignment, while those providing oversight at a lab or internship can be asked to help evaluate students. Students themselves may be asked to report on, interpret, or incorporate their on-site experience in the context of their other course activity.

Any prerequisites, such as a secured internship position or a prearranged work site, should be stated in your course description at registration time and in any catalogs that list your course, but should also be thoroughly explained in your syllabus. If a proctor or supervisor's report is required for fieldwork, this should also be clearly

indicated and available for students to download.

Assessment Considerations

As you design and develop your course, you will need to determine what types of assessment you'll use. We recommend you include both formative assessments and summative assessments.

Formative assessments are generally low stakes assessments that allow you to **monitor student learning**. They allow students to identify their strengths and weaknesses and you to recognize where students are struggling and address problems immediately through feedback, additional resources, instructional videos, etc.

Summative assessments are high-stakes assessments that allow you to **evaluate student learning** at the end of a unit or course. Examples of summative assessments include midterm exams, final research projects or papers, etc.

When designing your assessments, what matters most is that they are:

- relevant to the objectives and goals of the course
- appropriate for the level and scope of the course content
- easily enough accomplished online
- clearly outlined to students in the course schedule

It is also important that you vary your assessments to allow students to be assessed from a number of different perspectives, using a diverse set of skills.

Assignment Design

A poorly constructed assignment will negatively impact student success. It can be helpful to provide detailed assignment instructions to avoid student confusion. If you have instructional designers available to you, we encourage you to have them review your assignments to ensure they are clear from the student perspective. If you don't have these resources available, here are a few tips for writing better assignments for online courses.

Before faculty write up the instructions for an assignment, they should:

- Anticipate how their students will understand the assignment and actions

they are asked to take.

- Reproduce the steps and logical sequence and grouping of tasks associated with the assignment.

An assignment write-up should typically have the following components:

- **Overview:** A title and descriptive, introductory, or contextual portion that sets forth the background details that students need to know before attempting the assignment. For example, “One of the primary defining markers of the early modern age is a growing emphasis on the individual. This emphasis might take the form of an extended discussion of private romantic love, which is shown in the sonnets assigned for this week, or private perspective, which is evident in Montaigne's essays. It might take the form of a markedly pragmatic political philosophy, as you see in the selections from Machiavelli.”
- **Instructions:** Outline the main tasks or elements the assignment must cover or include, in logical order, along with directions for how and when to submit the assignment. The latter is not obvious in online classes. For example, “Select one of the readings assigned this week that emphasizes individualism. Write a 1,050- to 1,750-word paper analyzing three things: 1) How this author or work responded to earlier writers working on the same topics; 2) How this emphasis shaped the work's literary qualities; 3) What can current society learn from the work you discuss? Include examples or quotes to support or illustrate your assertions. Submit your paper as an attachment in Canvas.
- **Assessment Criteria:** You can list the criteria as a number or bulleted list or, even better, provide a grading rubric that outlines the performance criteria you will be using to assess the assignment. Keep in mind that students will use your assessment criteria as guidelines for completing their work, so put some careful thought into what you include and what you leave out. Include any definitions that might not be obvious to students. For example, if you haven't already explained what is meant by a “persuasive essay” or “position paper,” “peer reviewed articles,” “analytic,” or “substantive,” be sure to do so, and when appropriate, provide examples, models, or resources to assist students.

For more detail on writing up instructions, see the [Writing Up an Assignment and Using Rubrics](#) tutorial module from CUNY School of Professional Studies.

High-Stakes and Low-Stakes Testing

When designing your course, you will need to decide which types of quizzes or tests you want to use in your course and whether or not a specific test will be used for

self-assessment, review, and/or practice. You will also want to decide if the test is a low-stakes test or a high-stakes test.

Low-Stakes Testing. While low-stakes tests are not generally seen as an effective learning technique in face-to-face courses, research in online learning has shown they can be very effective in helping students to attain a grasp of factual information and that such tests enable the repeated practice needed for students to proceed to tackle more complex assignments that involve higher-level thinking. By building in automatic feedback, and allowing multiple attempts for no or little credit, retention of foundational material and a gain in confidence are promoted.

High-Stakes Testing. High-stakes tests are generally comprehensive, spanning multiple units or modules in a course. The purpose of these tests tends to be evaluative in nature and are used to capture how well students are able to apply what they have learned in a course.

Academic Integrity

Academic Integrity is just as important in the online environment as it is in the face-to-face environment.

Most institutions provide access to plagiarism detection services, like Turnitin or SafeAssign, and proctoring tools, like RPNOW or Respondus LockDown Browser. We encourage you to get familiar with these services and tools so you can interpret the results produced after a student submits their work or access the results after a student takes a test.

When designing quizzes or tests there are some strategies you can implement to ensure the integrity of the test.

- Create a large pool of questions from which to draw the test so each student has a different version of the test.
- Create questions that relate to all the materials in the course rather than just the textbook or the lectures.
- Use the testing features in your Learning Management System to set time limits to make it challenging for students to look up information and complete the test on time.
- Use a proctoring feature or lockdown browser if it is available at your school. Remote proctoring solutions can involve live monitoring, recording for later review, or other methods. These proctoring solutions are implemented on an

institution or system-wide basis and involve considerable costs on the part of either/or the institution and students, coordination, and planning.

Another strategy we recommend is taking a holistic approach to promoting academic integrity in your online course by emphasizing the concept of academic integrity rather than simply telling students “not to cheat.” You can provide a tutorial on academic integrity or simply have a discussion on what it means to maintain academic integrity in your online course.

An excellent publicly available resource on Academic Integrity for students can be found here:

Arizona State University: <https://thecollege.asu.edu/resources/academic-integrity>

Having a discussion with students or devising activities that cover academic integrity as part of the introductory materials in your course can help students get off on the right foot. Here are some strategies that Ko & Rossen (2017) mention in their book that encourages academic integrity through design (p. 80-82):

- Make sure that paper or essay topics are sufficiently specific, rather than so broad that any borrowed text could meet the requirements.
- Include an element that requires students to refer to discussions or other content specific to the class activity.
- When allowing for choice in picking topics, either provide a set of choices or require that students provide an outline, bibliography, and rough draft of their papers so that you can monitor the development of the work.
- Make sure that rubrics or listed criteria for grading the assignment include responsiveness to the assignment requirements and questions. While you may not be able to prove plagiarism, you can easily see what’s relevant and what is not.
- Ask students to give examples and evidence to support their arguments and theses.
- Insist on clear citations and a reference list that matches those citations.
- For group work, ask students to clearly label each person’s contribution.
- If students are creating multimedia products, ask them to read about fair use guidelines and document what they have borrowed from elsewhere.
- Use portfolios to measure student learning over time by a progressive series of work products. It isn’t difficult to assemble a portfolio of work in the online environment. The key is planning an adequate variety of activities from

which students can assemble portfolios of their work.

- Incorporate video chat into at least one high-stakes assessment. It can be the equivalent of an oral exam and then the grade is divided up between the oral and written portions. This can work especially well for a final project or paper.

Note: Learn more about the strategies that ASU Online suggests to maintain academic integrity in their booklet [Cheaters Never Prosper](#).